

A Randomized Controlled Trial of Preoperative Oral Supplementation With a Specialized Diet in Patients With Gastrointestinal Cancer

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Background & Aims: Perioperative nutrition with specialized enteral diets improves outcome when compared with standard formulas. A post-hoc analysis suggested preoperative administration as the most important period. Thus, we designed a study to understand prospectively whether preoperative supplementation could be as efficacious as the perioperative approach and superior to a conventional treatment (no artificial nutrition) in reducing postoperative infections and length of hospital stay. **Methods:** A total of 305 patients with preoperative weight loss <10% and cancer of the gastrointestinal tract were randomized to receive the following: (1) oral supplementation for 5 days before surgery with 1 L/day of a formula enriched with arginine, ω -3 fatty acids, and RNA, with no nutritional support given after surgery (preoperative group, $n = 102$); (2) the same preoperative treatment plus postoperative jejunal infusion with the same enriched formula (perioperative group, $n = 101$); and (3) no artificial nutrition before and after surgery (conventional group; $n = 102$). **Results:** The 3 groups were comparable for all baseline and surgical characteristics. Intention-to-treat analysis showed a 13.7% incidence of postoperative infections in the preoperative group, 15.8% in the perioperative group, and 30.4% in the conventional group ($P = 0.006$ vs. preoperative; $P = 0.02$ vs. perioperative). Length of hospital stay was 11.6 ± 4.7 days in the preoperative group, 12.2 ± 4.1 days in the perioperative group, and 14.0 ± 7.7 days in the conventional group ($P = 0.008$ vs. preoperative and $P = 0.03$ vs. perioperative). **Conclusions:** Preoperative supplementation is as effective as perioperative administration in improving outcome. Both strategies seem superior to the conventional approach.

Major surgeries for cancer of the gastrointestinal tract are still associated with a high rate of postoperative morbidity, which has elevated treatment costs and may consume a significant proportion of health care resources.^{1,2} Among the proposed strategies to reduce postoperative morbidity and related costs is the use of

enteral diets enriched with arginine, glutamine, ω -3 fatty acids, and other nutritional substrates (namely immunonutrition). These key nutrients seem to modulate the immune and inflammatory responses and gut function after surgery.³⁻⁶ Two recent randomized double-blind trials consistently suggested that perioperative (before and after surgery) administration of immunoenhancing diets could significantly decrease the incidence of infectious complications and length of hospital stay.^{7,8} The perioperative use of immunonutrition was also shown to be a predominant and most efficient strategy over standard enteral feeding.^{2,8} However, none of these studies were designed with separate randomization according to nutritional status of the patients. Moreover, in the study design, there were no control arms that received no nutritional support.⁹ In our previous trial, a post-hoc analysis showed that immunonutrition was effective not only in malnourished patients but also well-nourished patients.⁷ Similar results were obtained in a subgroup of patients who received only preoperative supplementation because they were not tolerant to postoperative early enteral infusion.¹⁰ Thus, we hypothesized that simple preoperative oral support with key nutrients could be as effective as the perioperative approach. Further, it may also have the advantage of avoiding the potential side effects of postoperative enteral nutrition. To validate this hypothesis, we designed a prospective randomized trial using well-nourished patients with the aim of comparing preoperative with perioperative treatment. A control group receiving a conventional treatment (no artificial feeding) was also evaluated.

The primary end point of the study was the reduction of postoperative infection rate and length of postoperative stay. The secondary end point was to investigate the

Abbreviations used in this paper: BMI, body mass index.

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effect of the different regimens on nutritional parameters, gut function, and patient compliance.

Patients and Methods

Between October 1998 and September 2000, we conducted a prospective randomized clinical trial at our institution. Inclusion criteria were histologically documented neoplasm of the gastrointestinal tract and planned major elective surgery. Exclusion criteria were weight loss $\geq 10\%$ (with respect to usual body weight) in the past 6 months, age younger than 18 years, hepatic dysfunction (Child–Pugh class $>B$), respiratory dysfunction (arterial $\text{PaO}_2 < 70$ torr), renal dysfunction (serum creatinine level > 3 mg/dL, hemodialysis), cardiac dysfunction (New York Heart Class > 3), Karnofsky score < 60 , pregnancy, ongoing infections, and immune disorder (neoadjuvant radiochemotherapy, circulating neutrophils $< 2.0 \times 10^9/\text{L}$). After applying these exclusion criteria, patients were allocated by individual random number into 3 arms. The randomization list was generated by a computer program. The study design was explained to potential participants, who then signed a consent form before randomization. None of the patients included in the present trial were reported in previous studies. The protocol was approved by the ethical committee of our hospital.

Before surgery, the first group of patients was asked to drink 1 L/day for 5 consecutive days of a supplemented liquid diet (Oral Impact; Novartis Consumer Health, Bern, Switzerland) (preoperative group). One liter of Oral Impact contains 12.5 g arginine, 3.3 g ω -3 fatty acids, and 1.2 g RNA. The kilocalorie/milliliter ratio is 1:1. In the postoperative course, the patients were given an intravenous solution of 5% glucose and electrolytes ($28 \text{ mL} \cdot \text{kg}^{-1} \cdot \text{day}^{-1}$) until the day of recovery of oral food.

The second group received the same preoperative treatment. In the postoperative period, these patients were given jejunal feeding with the same enriched formula starting within 12 hours after surgery (perioperative group). The enteral diet was administered via a feeding jejunostomy or a nasojejunal tube with a flow controlled by a peristaltic infusion pump. The initial rate was 10 mL/h and progressively increased by $20 \text{ mL} \cdot \text{h}^{-1} \cdot \text{day}^{-1}$ until reaching the daily volume of 1500 mL. The postoperative regimen was continued until patients resumed oral food.

The third group did not receive any artificial nutritional supplementation before surgery. In the postoperative course, the patients were given an intravenous solution of 5% glucose and electrolytes ($28 \text{ mL} \cdot \text{kg}^{-1} \cdot \text{day}^{-1}$) until the day of recovery of oral food (conventional group).

In both the preoperative and perioperative groups, oral supplementation was performed as outpatient therapy. All patients kept a written record of the amount of supplemented diet consumed daily. In the preoperative period, regardless of the type of treatment, all patients were advised to consume regular meals as desired. Hospital admission was scheduled for all patients 2 days before surgery.

Six days before surgery (day -6), the following baseline parameters were determined in all patients: body weight (kg), height (m), body mass index (BMI; body weight in kg/height in m^2), degree of weight loss (with respect to usual body weight in the previous 6 months), performance status according to the Karnofsky score, hemoglobin (g/L), plasma levels of total proteins (g/L), albumin (g/L), prealbumin (g/L), total circulating lymphocyte count ($\times 10^9/\text{mL}$), and arginine plasma level ($\mu\text{mol/L}$). Arginine level was also determined 1 day before surgery. Comorbidity factors were also recorded in all subjects. Patients with a BMI value between 18 and 25 were considered normal, between 25.1 and 30 considered overweight, and > 30 considered obese according to a previous definition.¹¹ Type and duration of surgery, operative blood loss, and rate and amount of homologous blood transfused were registered.

Intestinal washout with an isosmotic solution (3 L) was performed the day before surgery in patients undergoing colorectal surgery. Patients were also treated with an enema the evening before and the morning of surgery. These patients received antibiotic prophylaxis with a single intravenous dose (2 g) of cefotetan disodium 30 minutes before surgery. Patients scheduled for gastric, esophageal, or pancreatic surgery were treated with intestinal washout (1 L) the day before surgery and antibiotic prophylaxis with a single intravenous dose (2 g) of cefazolin sodium 30 minutes before surgery. A second dose of antibiotic was given if the duration of the surgical procedures was longer than 4 hours. Prophylaxis for deep vein thrombosis was performed by low-molecular-weight heparin (50 UI/kg).

Side effects of postoperative enteral feeding such as abdominal cramping, bloating, diarrhea (defined as more than 3 liquid stools/day), vomiting, and displacement or clogging of the feeding tube were evaluated in all patients. We also registered the date of the first flatus and bowel movement for all patients.

Trained members of the surgical staff who were not directly involved in the study were required to register postoperative complications as previously a priori defined.¹² They also assessed independently the indication and duration of antibiotic therapy in the postoperative course, the first day of oral food resumption, and the day of hospital discharge.

Follow-up for complications was performed during office visits for 30 days after hospital discharge.

Statistics

We assumed a postoperative infection rate of 30% in the conventional group.⁷ A reduction to 15% would be considered to indicate the efficacy of treatment. With a maximum 100 individuals in each group, we were about 80% powered to detect such a reduction at an α level of 0.05. All patients were analyzed on an intention-to-treat basis. Evaluation of the rate of complication in the subgroups stratified for BMI value and the duration of antibiotic therapy were performed as post-hoc analyses.

Descriptive results are shown as mean $\pm$ SD or number of patients (%). One-way analysis of variance and the Kruskal–

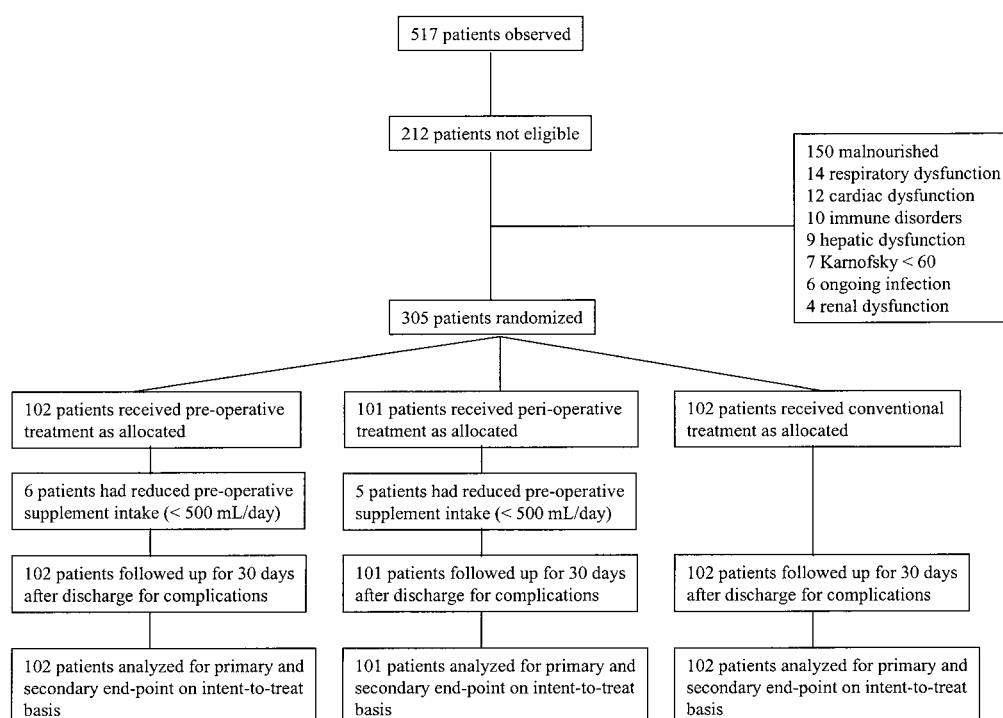


Figure 1. Diagram of the trial according to the CONSORT statement.

Wallis test (for not normally distributed data, i.e., operative time, operative blood loss, length of stay, resumption of oral food, and duration of antibiotic therapy) were used to compare continuous variables among groups. Post-hoc multiple comparisons were performed by the Bonferroni correction. The χ^2 test (with Yates' correction if there were cells with fewer than 10 observations) and Fisher exact test were used to compare discrete variables. All *P* values are 2 sided, and significance was set at *P* < 0.05. The SPSS package version 8.0 for Windows (SPSS Inc., Chicago, IL) was used for the statistical analysis.

Results

Figure 1 shows a diagram of the trial according to the CONSORT statement.¹³ During the study period, 517 patients were observed. Of these, 305 were judged eligible and entered by randomization into either the preoperative group (*n* = 102), perioperative group (*n* = 101), or conventional group (*n* = 102). All eligible patients were analyzed on an intention-to-treat basis.

The mean preoperative intake of the previously described oral supplementation was 925 mL/day without difference between the preoperative and perioperative groups.

Table 1 lists the preoperative (day -6) characteristics and the comorbidity factors of the eligible patients. The 3 study arms were well balanced for all parameters, even though the perioperative group had slightly more comorbidity factors than the other 2 groups (*P* = NS). One day before surgery, the arginine plasma level was 68.1 ± 12.4 $\mu\text{mol/L}$ in the conventional group, 101.2 ± 20.8 $\mu\text{mol/L}$ in the preoperative group (*P* = 0.002 vs. con-

ventional), and 98.8 ± 17.5 $\mu\text{mol/L}$ in the perioperative group (*P* = 0.005 vs. conventional). The 3 groups were also comparable for surgical variables (Table 2). Most patients underwent upper gastrointestinal tract surger-

Table 1. Preoperative Characteristics

	Conventional (<i>n</i> = 102)	Preoperative (<i>n</i> = 102)	Perioperative (<i>n</i> = 101)
Age (yr)	63.4 $\pm$ 11.9	62.3 $\pm$ 12.3	65.6 $\pm$ 11.5
M/F	56:46	50:52	60:41
Body weight (kg)	68.1 $\pm$ 11.7	69.4 $\pm$ 10.1	69.0 $\pm$ 13.3
BMI (kg/m ²)	23.8 $\pm$ 4.1	24.5 $\pm$ 4.9	24.2 $\pm$ 4.5
18-25	65	68	62
25.1-30	25	23	26
>30	12	11	13
Weight loss (%)	2.3 $\pm$ 2.7	2.4 $\pm$ 2.6	2.5 $\pm$ 2.7
Karnofsky score (%)	86 $\pm$ 11	84 $\pm$ 12	88 $\pm$ 10
Hemoglobin (g/L)	126 $\pm$ 13	123 $\pm$ 15	121 $\pm$ 12
Total proteins (g/L)	68 $\pm$ 6.1	68 $\pm$ 6.0	67 $\pm$ 7.1
Albumin (g/L)	40.3 $\pm$ 6.5	40.2 $\pm$ 5.6	39.9 $\pm$ 5.6
Prealbumin (g/L)	0.24 $\pm$ 0.07	0.26 $\pm$ 0.08	0.25 $\pm$ 0.06
Arginine ($\mu\text{mol/L}$)	65.7 $\pm$ 9.1	66.8 $\pm$ 11.3	64.2 $\pm$ 12.0
Lymphocytes ($\times 10^9/\text{mL}$)	1.68 $\pm$ 0.49	1.73 $\pm$ 0.65	1.62 $\pm$ 0.53
Comorbidity			
Hypertension	22	16	19
Diabetes	5	6	8
Arrhythmia	4	7	6
Liver disease	5	3	5
Gastrointestinal tract disease	2	3	5
Cardiac disease	2	2	2
Central nervous system disease	2	1	2
Miscellaneous	3	7	5

Table 2. Surgical Parameters

	Conventional (n = 102)	Preoperative (n = 102)	Perioperative (n = 101)
Gastroesophageal resections	44	48	46
Pancreatic resections	26	28	27
Colorectal resections	32	26	28
Operative time (min)	220 ± 90	226 ± 92	237 ± 107
Operative blood loss (mL)	435 ± 350	470 ± 370	520 ± 410
Transfused patients	32	34	37
Transfusion (mL)	495 ± 265	430 ± 160	550 ± 305

NOTE. Values are means ± SD or number of patients.

ies. The perioperative group had to some extent a greater blood loss, rate of transfusion, and amount of transfused blood than the other 2 groups, but these differences did not reach statistical significance.

Outcome variables are reported in Table 3. Overall postoperative mortality was 1.6% (5 of 305). No difference was observed among groups. The overall postoperative complication rate was 39% (119 of 305).

A significant reduction in the number of patients who developed postoperative infections was found in both the preoperative and perioperative groups compared with the conventional group. The number of patients with non-infectious or any complications was consistently lower in the preoperative and perioperative groups than in the conventional group. The difference, however, did not reach statistical significance. The rate of infectious, non-infectious, or any complications was similar between the preoperative and perioperative groups. The mean length of postoperative stay was significantly shorter in the patients receiving the supplemented formula (both preoperative and perioperative treatments) than in the conventional group. The post-hoc analysis of the mean duration of antibiotic therapy needed to treat postoperative

Table 3. Outcome Variables

	Conventional (n = 102)	Preoperative (n = 102)	Perioperative (n = 101)
Death	1	1	2
Patients with infectious complications	31	14 ^a	16 ^b
Patients with noninfectious complications	36	30	28
Patients with any complication	49	36	34
Length of hospital stay (days)	14.0 ± 7.7	11.6 ± 4.7 ^c	12.2 ± 4.1 ^d

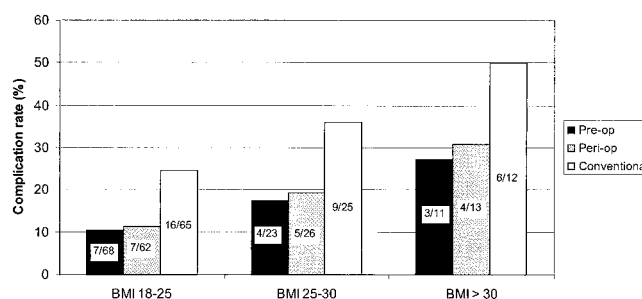
NOTE. Values are means ± SD or number of patients.

^a*P* = 0.006 vs. conventional.

^b*P* = 0.02 vs. conventional.

^c*P* = 0.008 vs. conventional.

^d*P* = 0.03 vs. conventional.

**Figure 2.** Rate of infectious complications in the 3 groups according to different BMI values.

infections showed 6.0 ± 2.6 days in the preoperative group versus 6.3 ± 2.9 days in the perioperative group and 9.2 ± 5.9 days in the conventional group (*P* = 0.01 vs. preoperative; *P* = 0.03 vs. perioperative).

The post-hoc analysis of postoperative complications split for different thresholds of BMI showed a rate of 35.9% (70 of 195) for BMI values between 18 and 25, 39.1% (29 of 74) for BMI values between 25.1 and 30, and 55.5% (20 of 36) for BMI values >30 (*P* = 0.04; BMI >30 vs. BMI 18–25). Postoperative infection rate according to BMI value and in the 3 arms is shown in Figure 2. Both preoperative and perioperative supplementation consistently reduced the incidence of infectious complications compared with conventional treatment in all 3 subgroups of patients, although none of these differences reached statistical significance.

Table 4 shows the complications in detail. We observed a trend toward a higher rate of wound infections, abdominal abscesses, and respiratory tract infections in the conventional group than in the other 2 groups but without statistical difference. The rate of anastomotic leak was 9.8% (10 of 102) in the conventional group,

Table 4. Detail of Complications

	Conventional (n = 102)	Preoperative (n = 102)	Perioperative (n = 101)
Wound infection	11	7	7
Abdominal abscess	10	4	4
Respiratory failure	6	6	9
Respiratory tract infection	8	3	6
Delayed gastric emptying	5	6	6
Pancreatic fistula	5	5	8
Urinary tract infection	5	4	2
Circulatory insufficiency	4	6	7
Bleeding	4	2	1
Bacteremia	5	0	1
Wound dehiscence	3	3	1
Pleural effusion	4	2	1
Renal dysfunction	2	1	0
Sepsis	2	0	1
Intestinal obstruction	1	1	0
Pulmonary embolism	0	0	1

3.9% (4 of 102) in the preoperative group, and 4.9% (5 of 101) in the perioperative group.

Patients receiving postoperative enteral feeding (perioperative group) had a faster recovery of bowel function than subjects who did not. In fact, the first flatus occurred after 2.8 ± 1.1 days in the perioperative group versus 3.5 ± 1.3 days and 3.6 ± 1.2 days in the preoperative and conventional groups, respectively ($P = 0.0001$). Moreover, the first bowel movement occurred after 4.3 ± 1.3 days in the perioperative group versus 5.4 ± 1.5 days and 5.2 ± 1.7 days in the preoperative and conventional groups, respectively ($P = 0.0001$). Recovery of oral food intake after surgery in the perioperative group was after 7.1 ± 3.2 days, in the preoperative group after 6.7 ± 2.7 days, and in the conventional group after 7.3 ± 3.8 days ($P = \text{NS}$). The patients receiving perioperative treatment experienced lower weight loss during the hospital stay than the other 2 groups. The mean weight loss (with respect to admission) at 8 days after surgery was 3.0% in the perioperative group, 5.2% in the preoperative group, and 4.8% in the conventional group.

In the perioperative group, clogging of the feeding tube was recorded in 3 patients and displacement of the nasojejun tube in 6 patients. Postoperative abdominal cramping/bloating was observed in 42 patients receiving enteral feeding and in 16 and 14 patients in the preoperative and conventional groups, respectively ($P < 0.0001$ vs. perioperative group). There were 7 patients with diarrhea in the perioperative group, 3 in the preoperative group, and 3 in the conventional group ($P = \text{NS}$). There were 2 patients with vomiting in the perioperative group, 1 in the preoperative group, and 2 in the conventional group ($P = \text{NS}$). Despite these side effects, the overall rate of tolerance to postoperative enteral infusion was 88.1% (89 of 101).

Discussion

Systematic reviews of the literature and international guidelines suggest that elective surgical patients should receive artificial nutritional support only if a preoperative weight loss $\geq 10\%$ is detected or if adequate recovery of oral food intake is expected later than 10 days after surgery.^{14–19} Recently, a multicenter trial showed that malnourished patients fed enterally had significantly less postoperative complications than parenterally fed subjects.¹² It should be noted that the above indication to artificial nutrition is valid if considering the standard composition of the formulas, prescribed with the primary aim of providing calories and proteins. Yet, in past years, standard nutritional feeds have been modified by adding

specific substrates, such as arginine, glutamine, ω -3 fatty acids, and others. In the experimental and clinical setting, immunonutrition has been shown to up-regulate host immune response, control inflammatory response, modulate the synthesis of acute-phase and constitutive proteins, improve gut oxygenation and barrier function after injury, and reduce septic morbidity and mortality.^{3–6,20–25} As a consequence, these new substrates should not be given with the aim of providing calories and nitrogen but rather to modulate the host response and counteract the postoperative immunometabolic alterations induced by surgical injury which, per se, may increase patient susceptibility to postoperative complications. With this rationale, immunonutrition may also be indicated in well-nourished patients.

Three meta-analyses have shown that postoperative use of enteral immunonutrition may improve outcome when compared with standard feeds.^{26–28} Nevertheless, by reviewing the studies with a large sample size,^{3,4,29} the results are not always consistent. Moreover, the potential harm of the control diets instead of the benefit of immunoenhancing formulas could not be excluded for the lack of unfed groups. In fact, in all of the above trials, enteral immunonutrition was always compared with isoenergetic and isocaloric standard enteral diets. Only one study, by Heslin et al.³⁰ (not included in the meta-analyses), was designed to compare immunonutrition with fluid and electrolyte support. This trial reported a similar rate of complications in the treated and control groups, suggesting that the simple postoperative provision of key nutrients was not effective on relevant outcome variables. One potentially confounding factor was that the patients randomized to immunonutrition received a very small amount of key substrates (approximately 30% of the planned goal in the first week). Thus, it may be that the lack of beneficial effects was partially related to the limited dose given. Overall, the data on postoperative immunonutrition suggest a more general consideration: if several days of feeding and an adequate dose of key nutrients is not provided, neither a significant improvement of the host defense mechanisms nor a reduction of complications can be achieved. As with any other substance with supposed pharmacologic action, immunonutrients should reach suitable tissue and plasma concentration to be active. For this reason, we are convinced that a key point would be to anticipate the provision of immunonutrients before surgery to obtain adequate levels at the time of surgical stress when the need for stimulation of the immune system is maximized.

Two recent large, randomized, double-blind trials^{7,8} consistently suggested that perioperative administration of immunoenhancing diets could significantly decrease the incidence of infectious complications and the length of hospital stay when compared with isoenergetic and isonitrogenous feeds. Based on our results of phase 2 trials,^{31,32} we speculated that the reduction of postoperative infections was related to the immunologic and metabolic advantages found in patients perioperatively supplemented with immunonutrition. The perioperative administration of an enteral diet enriched with arginine, ω -3 fatty acids, and RNA prevented the early postoperative decrease of both the phagocytosis ability and the respiratory burst of polymorphonuclear cells; significantly increased the nitric oxide plasma concentration, cell-mediated immune response, plasma levels of interleukin 2, and CD4/CD8 ratio immediately after surgery; reduced the plasma levels of C-reactive protein and interleukin 6; and improved both intestinal microperfusion and postoperative gut mucosal oxygen metabolism.

A subsequent post-hoc analysis of our data⁷ suggested that perioperative immunonutrition was efficacious regardless of the baseline nutritional status of the patients. In fact, perioperative administration of immunoenhancing diets reduced postoperative infections both in well-nourished and malnourished patients. Another post-hoc analysis showed it was possible to improve outcomes in a subgroup of patients who received only preoperative supplementation because of intolerance to postoperative early enteral infusion.¹⁰ It was argued that a potential limitation of this type of trial was the absence of a group of patients without any nutritional support.⁹ For these reasons, we decided to undertake a new study with a double aim: to prospectively compare preoperative and perioperative treatment and to understand whether immunonutrition is superior to a conventional treatment (no artificial feeding). To include a nonfed control group, we had to exclude malnourished patients because we considered it unethical to restrain any form of artificial nutritional support in these patients.

The present data suggest that, in well-nourished patients, the administration of specific substrates may improve selected outcome surgical variables when compared with a conventional treatment. These results are the first of this kind and are difficult to compare with other trials because of the study design. In fact, no other report compared preoperative and perioperative immunonutrition with no feeding. In contrast, there are numerous data comparing no nutrition versus parenteral nutrition in elective surgical patients who are well nourished or moderately malnourished. The pooled results do not

support the routine use of nutritional support in this cohort of patients.^{18,19} Moreover, large well-designed trials reported potential detrimental effects of parenteral overfeeding in well-nourished subjects.^{33–35} Different results come by the comparison of early postoperative enteral feeding with postoperative fasting. In fact, a systematic review by Lewis et al. suggests that there is no advantage to keeping patients “nil by mouth” after elective gastrointestinal resection.³⁶

As discussed, the only trial comparing an immunoenhancing formula with no feeding was designed to provide the diet only in the postoperative course.³⁰ These results allow us to reinforce our hypothesis that preoperative supplementation is the key factor and that these new substrates protect the host and improve outcomes by a pharmacologic rather than by a nutritional mechanism, provided an adequate dose is administered.

Our positive results are not consistent with another recent trial³⁷ on preoperative immunonutrition. However, the sample size of the above study was quite small, the amount of diet intake was limited, and no modification of immunometabolic parameters was observed, rendering the conclusions not completely reliable. Another randomized, controlled trial³⁸ performed in 49 malnourished patients with head and neck cancer showed that perioperative supplementation with arginine alone did not affect nutritional, immunologic, and outcome variables. Thus, it may be speculated that the combination of more substrates could be more effective.

Furthermore, our results showed that, in this cohort of patients, simple preoperative supplementation with immunonutrition was sufficient and that this may represent the critical period to interact and modulate the host response. In fact, we observed that the postoperative continuation of enteral immunonutrition was useless and did not offer further advantages on reduction of morbidity when compared with simple preoperative treatment. Conversely, by avoiding postoperative enteral infusion, it was possible to reduce significantly gastrointestinal side effects, thereby improving patient compliance during the postoperative course. Also, it should be noted that it is possible to save health care resources by giving only preoperative immunonutrition. In fact, the daily cost of postoperative immunonutrition is approximately 150 euro per patient.^{2,8}

Despite the fact that postoperative enteral immunonutrition has a well-established rate of gastrointestinal side effects and costs,^{2,4,7,29} this form of feeding might be useful especially in a malnourished population. We reported recently in a similar randomized, controlled trial³⁹ that perioperative treatment resulted in the best

outcome in 150 malnourished subjects. This may be explained by the fact that undernourished patients, in addition to experiencing an impairment of the postoperative immune response, have increased energy and nitrogen requirements. Therefore, a continuation of immunonutrition in the postoperative course may be beneficial.

A post-hoc analysis of the present trial suggests that, in the general cohort of well-nourished patients, there are 3 subgroups of patients according to BMI. Subjects with a BMI value between 18 and 25 have a low risk of morbidity, with a progressively increased risk of postoperative complications in overweight and obese patients. These results fully confirm previous experiences showing obesity as a major negative predictive variable on surgical outcome.⁴⁰ Our results also suggest that immunonutrition reduces the incidence of postoperative infections in all 3 subgroups of normal, overweight, and obese subjects with a similar pattern.

In conclusion, in our population, the administration of a supplemented diet significantly improved outcome compared with a conventional treatment. The simple preoperative supplementation was as effective as the perioperative approach. Additionally, it could reduce gastrointestinal side effects. Furthermore, we suggest in the analysis of future nutritional trials that surgical patients be stratified per BMI. This will enable a better evaluation of the risk of postoperative complications and the impact of different treatments.

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